

Laplace Transformation

Lecture 07 - Detailed Notes, Corrected Formulas, Worked Solutions and ThermoGuard Case Study

A. Corrections Found in the Handwritten Notes and Case Study

Every item below was checked line by line. Wrong items in the original are listed with the correct version and the reason.

#	Where	As written	Correct	Why
1	Definition	$\mathcal{L}\{F(t)\}$	$\mathcal{L}\{f(t)\} = F(s)$	$f(t)$ is the time function; $F(s)$ is its transform. They are different letters.
2	Exponential	$\mathcal{L}\{e^{kt}\} = 1/e^{s+k}$	$1/(s - k)$	The answer is an algebraic fraction in s . No exponential stays, and the sign is $s - k$.
3	Exponential	$\mathcal{L}\{e^{2t}\} = 1/e^{s+2}$	$1/(s - 2)$	Same rule with $k = 2$.
4	Exponential	$\mathcal{L}\{e^{-2t}\} = 1/e^{s-2}$	$1/(s + 2)$	$k = -2$ gives $s - (-2) = s + 2$.
5	Question 2	$\mathcal{L}\{e^{-t}\} = 1/e^{s-1}$	$1/(s + 1)$	$k = -1$ gives $s + 1$.
6	Hyperbolic	$\mathcal{L}\{\sinh 2t\} = 2/(s^2 + 4)$	$2/(s^2 - 4)$	The formula $k/(s^2 - k^2)$ was right; the example used a plus sign.
7	Hyperbolic	$\mathcal{L}\{\cosh 4t\} = 4/(s^2 - 16)$	$s/(s^2 - 16)$	$\cosh$ has s in the numerator, not k .
8	Shift working	$s \rightarrow s \rightarrow k$	$s \rightarrow s - k$	Typo in the replacement rule.
9	Case study	$\mathcal{L}\{dT/dt\} + a\mathcal{L}\{T\}$ (no right side)	$\dots = b\mathcal{L}\{U(t)\}$	The input term $bU(s)$ was missing.
10	Case study	$T(t)$ and $T(s)$ both called T	$\Theta(s) = \mathcal{L}\{T(t)\}$	One letter for two different functions is confusing.
11	Case study	'predicted whether overshoot would occur'	A first-order model never overshoots	The solution is a pure exponential, so there are no oscillations (see Case Study, Section 5).
12	Case study	'reach stable temperature in 6 minutes'	Settles to within tolerance in about 6 min	e^{-at} never becomes exactly 0, so settling time (for example 2%) is used.
13	Case study	'Laplace model learned over time'	Parameters a , b are re-estimated	The transform does not learn. Estimating a and b from data (system identification) does.

#	Where	As written	Correct	Why
14	Case study	'unstable response' via Laplace	Pole at $s = -a$ moves to $s > 0$	Instability means a pole in the right half-plane, i.e. $a < 0$.

Everything else in the notes was checked and is **correct**: $\mathcal{L}\{1\}$, $\mathcal{L}\{n\}$, $\mathcal{L}\{t^n\}$, $\mathcal{L}\{\sin kt\}$, $\mathcal{L}\{\cos kt\}$, $\mathcal{L}\{\sinh kt\}$, $\mathcal{L}\{\cosh kt\}$ (as general formulas), the polynomial questions, the First Shift Theorem working, the derivative formulas, and the $f(t) = 3t^2$ derivative question.

1. Definition

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt = F(s)$$

The transform takes a function of time t (for $t \geq 0$) and gives a function of the new variable s . The integral only converges for large enough s , so each formula has a condition such as $s > 0$ or $s > k$.

Key properties

- **Linearity:** $\mathcal{L}\{a f(t) + b g(t)\} = a F(s) + b G(s)$. This lets us transform a sum term by term and pull constants out.
- **Zero:** $\mathcal{L}\{0\} = 0$.
- **Constant multiple:** $\mathcal{L}\{6t\} = 6 \cdot \mathcal{L}\{t\}$.

Deriving the basic formulas

$$\mathcal{L}\{1\}: \int_0^{\infty} e^{-st} dt = [-e^{-st}/s]_0^{\infty} = 0 - (-1/s) = 1/s \quad (s > 0)$$

$$\mathcal{L}\{e^{kt}\}: \int_0^{\infty} e^{-(s-k)t} dt = 1/(s - k) \quad (s > k)$$

$$\mathcal{L}\{t\}: \text{by integration by parts, } \int_0^{\infty} t e^{-st} dt = 1/s^2$$

2. Basic Formulas (Corrected)

Constant function

$f(t)$	$\mathcal{L}\{f(t)\} = F(s)$	Condition
1	$1/s$	$s > 0$
2	$2/s$	$s > 0$
3	$3/s$	$s > 0$
n (a constant)	n/s	$s > 0$

Algebraic functions

$f(t)$	$\mathcal{L}\{f(t)\}$
t	$1!/s^2$
t^2	$2!/s^3$
t^3	$3!/s^4$
t^n	$n!/s^{n+1} \quad (n = 0, 1, 2, \dots)$

Exponential functions (corrected)

$f(t)$	$\mathcal{L}\{f(t)\}$	Condition
e^{kt}	$1/(s - k)$	$s > k$
e^{2t}	$1/(s - 2)$	$s > 2$
e^{-2t}	$1/(s + 2)$	$s > -2$

$f(t)$	$\mathcal{L}\{f(t)\}$	Condition
e^{-t}	$1/(s + 1)$	$s > -1$

Memory trick: $\mathcal{L}\{e^{kt}\} = 1/(s - k)$. To find the result, take the number multiplying t in the exponent and subtract it from s . A negative k turns the minus into a plus.

3. Trigonometric and Hyperbolic Formulas

#	Function	Formula	Example
1	$\mathcal{L}\{\sin kt\}$	$k/(s^2 + k^2)$	$\mathcal{L}\{\sin 2t\} = 2/(s^2 + 4)$
2	$\mathcal{L}\{\cos kt\}$	$s/(s^2 + k^2)$	$\mathcal{L}\{\cos 4t\} = s/(s^2 + 16)$
3	$\mathcal{L}\{\sinh kt\}$	$k/(s^2 - k^2)$	$\mathcal{L}\{\sinh 2t\} = 2/(s^2 - 4)$
4	$\mathcal{L}\{\cosh kt\}$	$s/(s^2 - k^2)$	$\mathcal{L}\{\cosh 4t\} = s/(s^2 - 16)$

Why sinh and cosh have a minus sign. Since $\sinh kt = (e^{kt} - e^{-kt})/2$:

$$\mathcal{L}\{\sinh kt\} = \frac{1}{2}[1/(s - k) - 1/(s + k)] = \frac{1}{2} \cdot 2k/(s^2 - k^2) = \mathbf{k/(s^2 - k^2)}$$

$$\mathcal{L}\{\cosh kt\} = \frac{1}{2}[1/(s - k) + 1/(s + k)] = \frac{1}{2} \cdot 2s/(s^2 - k^2) = \mathbf{s/(s^2 - k^2)}$$

Sin and cos use the same idea with complex exponentials, which gives the plus sign $s^2 + k^2$.

Rule of thumb: sin and cos $\rightarrow$ **plus** in the denominator. sinh and cosh $\rightarrow$ **minus**. sin and sinh have **k** on top; cos and cosh have **s** on top.

4. Solved Questions (Step by Step)

Question 1: $f(t) = t^2 + 6t + 17$

Step 1 - split by linearity: $\mathcal{L}\{f(t)\} = \mathcal{L}\{t^2\} + 6\mathcal{L}\{t\} + 17\mathcal{L}\{1\}$

Step 2 - apply formulas: $= 2!/s^3 + 6 \cdot (1!/s^2) + 17 \cdot (1/s)$

Step 3 - simplify: $= 2/s^3 + 6/s^2 + 17/s$

Step 4 - common denominator s^3 : $= (2 + 6s + 17s^2) / s^3$

Question 2: $f(t) = 4t^4 + 3t^3 + 2t^2 + t + 10$

Step 1: $\mathcal{L}\{f(t)\} = 4\mathcal{L}\{t^4\} + 3\mathcal{L}\{t^3\} + 2\mathcal{L}\{t^2\} + \mathcal{L}\{t\} + 10\mathcal{L}\{1\}$

Step 2: $= 4 \cdot 4!/s^5 + 3 \cdot 3!/s^4 + 2 \cdot 2!/s^3 + 1!/s^2 + 10/s$

Step 3 - factorials: $4! = 24, 3! = 6, 2! = 2, 1! = 1$

Step 4: $= 4 \cdot 24/s^5 + 3 \cdot 6/s^4 + 2 \cdot 2/s^3 + 1/s^2 + 10/s$

Answer: $96/s^5 + 18/s^4 + 4/s^3 + 1/s^2 + 10/s$

Questions on the trigonometric and exponential formulas

#	Question	Working	Answer
1	$\mathcal{L}\{\sin 3t\}$	$k = 3$ in $k/(s^2 + k^2)$	$3/(s^2 + 9)$
2	$\mathcal{L}\{e^{-t}\}$	$k = -1$, so $s - (-1)$	$1/(s + 1)$

Question 3: $f(t) = \cos 2t + t^2 + \sinh 3t + 1$

$\mathcal{L}\{f(t)\} = \mathcal{L}\{\cos 2t\} + \mathcal{L}\{t^2\} + \mathcal{L}\{\sinh 3t\} + \mathcal{L}\{1\}$

• $\cos 2t$: $k = 2 \rightarrow s/(s^2 + 4)$

• $t^2 \rightarrow 2!/s^3 = 2/s^3$

• $\sinh 3t$: $k = 3 \rightarrow 3/(s^2 - 9)$

• $1 \rightarrow 1/s$

Answer: $s/(s^2 + 4) + 2/s^3 + 3/(s^2 - 9) + 1/s$

5. First Shift Theorem

When a function is multiplied by an exponential e^{kt} , use the First Shift Theorem:

$$\mathcal{L}\{e^{kt} \cdot f(t)\} = F(s - k)$$

In words: find $F(s)$ for the other part first, then replace every s with $(s - k)$.

Example: $\mathcal{L}\{e^{2t} \cdot \sin 3t\}$

Step 1: $\mathcal{L}\{\sin 3t\} = 3/(s^2 + 9) = F(s)$

Step 2: here $k = 2$, so replace s by $(s - 2)$

Step 3: $3/((s - 2)^2 + 9) = 3/(s^2 - 4s + 4 + 9) = 3/(s^2 - 4s + 13)$

Questions

#	Question	$F(s)$	k	Answer
1	$\mathcal{L}\{e^{2t} \cdot t\}$	$1/s^2$	2	$1/(s - 2)^2$
2	$\mathcal{L}\{e^{-t} \cdot t\}$	$1/s^2$	-1	$1/(s + 1)^2$

Question 3: $f(t) = e^{2t}(\cos 2t + t^2 + \sinh 3t + 1)$

Step 1: transform the bracket: $F(s) = s/(s^2 + 4) + 2/s^3 + 3/(s^2 - 9) + 1/s$

Step 2: replace every s by (s - 2):

Answer: $(s - 2)/((s - 2)^2 + 4) + 2/(s - 2)^3 + 3/((s - 2)^2 - 9) + 1/(s - 2)$

6. Laplace Transform of Derivatives

These formulas turn derivatives into multiplication by s , with the initial conditions included automatically.

Derivative	$\mathcal{L}\{ \cdot \}$
$f'(t)$	$sF(s) - f(0)$
$f''(t)$	$s^2F(s) - s f(0) - f'(0)$
$f'''(t)$	$s^3F(s) - s^2 f(0) - s f'(0) - f''(0)$
$f^{(n)}(t)$	$s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - f^{(n-1)}(0)$

Where the first formula comes from

$\mathcal{L}\{f'(t)\} = \int_0^\infty e^{-st} f'(t) dt$. Integration by parts gives $[e^{-st} f(t)]_0^\infty + s \int_0^\infty e^{-st} f(t) dt = 0 - f(0) + sF(s)$.

Question: $f(t) = 3t^2$

Step 1: $F(s) = \mathcal{L}\{3t^2\} = 3 \cdot 2! / s^3 = 6/s^3$

Step 2: $f(0) = 3(0)^2 = 0$

Step 3: $\mathcal{L}\{f'(t)\} = sF(s) - f(0) = s \cdot (6/s^3) - 0 = 6/s^2$

Check: $f(t) = 6t$ directly, and $\mathcal{L}\{6t\} = 6 \cdot (1/s^2) = 6/s^2$. The two methods agree.

7. Quick Reference Sheet

$f(t)$	$F(s)$
1 (or constant c)	$1/s$ (c/s)
t^n	$n! / s^{n+1}$
e^{kt}	$1/(s - k)$
$\sin kt$	$k/(s^2 + k^2)$
$\cos kt$	$s/(s^2 + k^2)$
$\sinh kt$	$k/(s^2 - k^2)$
$\cosh kt$	$s/(s^2 - k^2)$
$e^{kt} f(t)$	$F(s - k)$
$f'(t)$	$sF(s) - f(0)$
$f''(t)$	$s^2F(s) - s f(0) - f'(0)$

Case Study: ThermoGuard

Smart Temperature Control & Stability Prediction System

Topic: Application of Laplace Transformation

1. Problem

At Riphah International University, a software engineering student named **Kashaf Fatima** was working late in the computer lab. During summer, she noticed a common issue in university classrooms and server rooms: the temperature control systems reacted **too slowly**. Sometimes the room became too hot before the AC responded; other times it became too cold due to over-cooling. These sudden changes:

- damaged sensitive servers
- wasted electricity
- made people uncomfortable

Kashaf asked a simple question:

“Can a software system predict temperature behavior instead of reacting too late?”

That question led to the creation of **ThermoGuard**.

2. Building the System

ThermoGuard was a **software-based intelligent temperature monitoring system**. It consisted of:

- A **temperature sensor** (T) to measure room temperature
- A **control software** running on a microcontroller
- A **display panel** showing temperature trends
- An **alert system** for instability warnings

The real power of ThermoGuard was not the hardware. It was the **mathematics inside the software**.

3. The Math Inside the System

Kashaf modeled the temperature system using a **first-order differential equation**:

$$dT(t)/dt + a \cdot T(t) = b \cdot U(t)$$

- $T(t)$ = room temperature
- $U(t)$ = cooling/heating input
- a, b = system constants ($a > 0$ for a stable room)

She realized that solving this repeatedly in real time was difficult and that the system also needed **initial temperature conditions**. That is when she used the **Laplace Transformation**.

4. How ThermoGuard Works (Laplace Logic)

Step 1 - transform every term (corrected: the right side was missing in the original):

$$\mathcal{L}\{dT/dt\} + a \cdot \mathcal{L}\{T(t)\} = b \cdot \mathcal{L}\{U(t)\}$$

Step 2 - use the derivative formula. Let $\Theta(s) = \mathcal{L}\{T(t)\}$ (the original used $T(s)$, the same letter as $T(t)$):

$$s\Theta(s) - T(0) + a\Theta(s) = bU(s)$$

Step 3 - solve algebraically for $\Theta(s)$:

$$\Theta(s) = [T(0) + bU(s)] / (s + a)$$

Why Laplace was important:

- Differentiation became multiplication by s
- The initial temperature $T(0)$ was automatically included
- The response could be predicted before changes happened

Correction note: the original page stopped at the left-hand side and wrote $T(s)$ for the transform. Both are fixed above: the right side $bU(s)$ was added, and the transform is written $\Theta(s)$.

5. The First Test (Worked Solution)

ThermoGuard was tested in the university server room. Initial conditions: **$T(0) = 30^\circ\text{C}$** , desired temperature = **22°C** .

Here is the inverse transform that the case study mentions but does not show. Assume a constant control input $U(t) = U_0$, so $U(s) = U_0/s$, and that it is tuned so that the steady-state temperature is $bU_0/a = 22^\circ\text{C}$.

Step 1: $\Theta(s) = T(0)/(s + a) + bU_0/[s(s + a)]$

Step 2 - partial fractions: $bU_0/[s(s + a)] = (bU_0/a)[1/s - 1/(s + a)]$

Step 3 - inverse transform using $\mathcal{L}\{1\} = 1/s$ and $\mathcal{L}\{e^{-at}\} = 1/(s + a)$:

$$T(t) = T(0)e^{-at} + (bU_0/a)(1 - e^{-at}) = 22 + (30 - 22)e^{-at} = 22 + 8e^{-at}$$

The room cools smoothly from 30°C toward 22°C . Because this is a pure decaying exponential:

- **No overshoot** is possible (corrected: the original lists overshoot as something to predict, but a first-order model has none).
- **Settling time:** the gap is $8e^{-at}^\circ\text{C}$. For example, with $a = 2/3$ per minute (an assumed value), at $t = 6$ min the gap is $8e^{-4} \approx 0.15^\circ\text{C}$, i.e. within about 2% of the starting gap. That matches the displayed message “Temperature will normalize in 6 minutes”. The temperature approaches 22°C but never reaches it exactly, so “normalize” means “within tolerance”.
- **Cooling power** can be adjusted by changing U_0 or a , which moves the final value or speed.

ThermoGuard displayed: **“System Stable. Temperature will normalize in 6 minutes.”**

6. Learning and Adapting

ThermoGuard continuously monitored temperature changes, rate of change, and system response time. Over time, the system learned which rooms heat up faster, which environments need slow cooling, and how to avoid sudden temperature shocks.

Correction note: the Laplace transform itself does not learn. What changes is the model's parameters a and b , which are re-estimated from measured data (system identification) in each cycle. The updated model is then analyzed with Laplace methods. A room that heats faster has a larger a or b .

7. The Emergency Alert

One afternoon the temperature was rising slowly, but the **rate of change** increased sharply. The analysis showed the response was becoming unstable and that the cooling delay could cause overheating.

Math behind it: the pole of $\Theta(s)$ is at $s = -a$. If $a > 0$, the pole is on the left (stable): the e^{-at} term decays. If a failing cooling unit pushes the effective a below 0, the pole moves to $s = -a > 0$ (right half-plane) and the term e^{-at} grows without limit, so the temperature runs away. The system triggered: **“Warning! Predicted temperature instability in 5 minutes.”** Technicians found a failing cooling unit, and damage was avoided.

8. The Future of Smart Control Systems

- ThermoGuard was proposed for smart buildings
- Data centers adopted similar models
- Energy consumption was reduced

Kashaf concluded:

“Laplace Transformation allows software systems to understand not just the present, but the future behavior of dynamic systems.”

9. Classroom Reflection (For Students)

1. How does the Laplace Transform simplify differential equations in control systems? *(Derivatives become multiplication by s , so the equation becomes algebra.)*
2. Why is handling initial conditions important in real systems? *(The room starts at 30 °C; ignoring $T(0)$ gives the wrong response.)*
3. Can Laplace-based models be used for traffic or network load systems? *(Yes, for any system whose behavior follows linear differential equations, e.g. queues or load that rises and falls.)*

Conclusion

Laplace Transformation helps software-based control systems analyze stability and predict system behavior by converting differential equations into algebraic form.